



INTI
International College Subang

UNIT DETAILS:

Unit code	COS30045
Unit name	Data Visualization
Lecturer	Rishma Fathima Basher
Assignment	Beyond the Fine: Visualising the Disparities in Australian Road Enforcement Actions
Tutorial Day Time	Monday - Friday 6:00pm to 10:00pm
Section	C1
Session	March 2026
Semester	1
Word Count	2902

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1.0 Introduction

1.1 Background & Motivation

Across Australia, road traffic accidents continue to be a significant public health and economic concern. Identifying the most effective methods to reduce these crashes requires a rigorous, evidence-based approach to enforcement. The Bureau of Infrastructure and Transport Research Economics (BITRE) has maintained comprehensive records of road safety enforcement activities such as speeding, seatbelt non-compliance, mobile phone usage, and drug testing since 2008.

Historically, this data was published in a highly aggregated format. However, to account for the complex demographic and geographic characteristics driving road infractions, BITRE updated its collection methodology in 2023. The dataset now features fine-grained, multi-dimensional variables, including precise geographical typologies, age group breakdowns, and crucially, the distinction between monetary fines, direct arrests, and formal criminal charges stemming from the same offence.

A significant drawback of BITRE's current official Police Enforcement Dashboard is its reliance on summary-level data, which fails to reflect this new structural depth. This oversight masks critical public safety narratives. Academic research has long highlighted the tension between automated enforcement and manual police intervention; while automated cameras are highly effective at mass compliance, they are often critiqued by the public as revenue-generating rather than safety-focused, whereas manual policing is proven to be essential for intercepting high-risk behaviours like unlicensed driving or intoxication (De Pauw et al., 2014; WHO, 2023). Without visualising these discrepancies, policymakers cannot accurately assess whether automated cameras are functioning primarily as revenue streams, or if police presence is effectively deterring severe criminal traffic behaviour.

Target Audience:

- **Primary:** Traffic policymakers, law enforcement strategists, and road safety analysts. They require a diagnostic tool to evaluate the balance between automated enforcement and targeted police deployment.
- **Secondary:** The public and data journalism organizations focused on government accountability. They require transparent, interactive storytelling to understand who is being enforced against, where, and how.

Importance of the Visualization:

“Enforcement opacity” is a problem this project aims to solve. Existing dashboard tools provide only an aggregated total penalty count. Through this data collection and visualization, the data will be provided with context, by identifying the severity of the outcome (fine or arrest). This is very important for determining if the automated cameras are being used, primarily as a revenue generating tool and whether manual intervention by police officers is adequate for dealing with high-risk behaviours and repeat offenders.

Key User Tasks:

- Contrast the volume of automated camera detections against actual police-issued actions (arrests/charges).
- Identify specific demographic age groups most prone to receiving escalated enforcement (arrests) compared to monetary fines.
- Investigate geographic policing discrepancies (e.g., Major Cities vs. Remote areas).
- Examine temporal trends to determine seasonal or longitudinal shifts in enforcement strategies.

1.2 Visualization Purpose

This visualization is designed to answer the following core questions, translating raw enforcement logs into actionable intelligence:

i. Severity Distribution:

For severe offences (e.g., unlicensed driving), what proportion results in standard fines versus arrests/charges? This reveals whether states rely solely on financial penalties or actively pursue criminal deterrence.

ii. Demographic Targeting:

Are younger demographics (17-25) over-represented in police-issued offences compared to automated camera detections, indicating targeted policing of high-risk age groups?

iii. Geographic & Methodological Discrepancy:

Does the primary detection method differ between 'Major Cities' and 'Remote' areas? (e.g., Is there a heavier reliance on manual police actions in remote areas lacking camera infrastructure?).

iv. Decision-Making Impact:

By transforming raw traffic logs into an interactive display, this dashboard provides policymakers with actionable intelligence for resource allocation, such as deploying police patrols to areas with low camera coverage. Crucially, visualizing arrests alongside fines shifts the policy narrative away from revenue generation and toward genuine crime deterrence.

Benefits and Decision-Making Impact:

By explicitly visualizing “Arrests” and “Charges” alongside “Fines”, this dashboard actively shifts the policy narrative away from revenue generation and toward genuine crime deterrence. Policymakers can use this interface to justify resource allocation, for instance, identifying regions where low camera coverage necessitates increased police patrols, it is ultimately supporting more effective, behaviour-focused road safety strategies.

2.0 Data

2.1 Data Source and Governance

Original Data Source:

The dataset utilized in this project is the “**Police Enforcement, Australia, 2024 (Fines)**” dataset, directly sourced from the official Australian Government Department of Infrastructure and Transport (BITRE) Data Hub.

Web Link:

https://datahub.roadsafety.gov.au/sites/default/files/documents/police_enforcement_2024_fines.xlsx

Data Summary Table:

Table 1: Data Summary Table for Police Enforcement, Australia, 2024 (Fines)

Attribute	Description	Data Type	Example Value
Total Records	Total granular enforcement records in the file	Count	12,179 rows
Key Dimensions	Dimensions added in BITRE's 2023 update	Categorical / Nominal	JURISDICTION, LOCATION, AGE_GROUP, METRIC, DETECTION_METHOD
Key Metrics	Quantitative outcomes of enforcement	Ratio / Quantitative	FINES, ARRESTS, CHARGES
Time Frame	The specific period covered by this dataset	Interval	January 2024 (Monthly snapshot)
Update Frequency	How often this specific granular data is released	Temporal	Annually

Data Collection & Quality Assessment:

The data is collected by state and territory police forces (e.g., NSW Police, VIC Police) and aggregated by BITRE to provide a standardized national view. Initial profiling via the KNIME Statistics Node revealed zero traditional NULL values. However, severe data quality issues presented themselves through mislabelled categorical values (e.g., Tasmania aggregates fixed and mobile camera data while other jurisdictions disaggregate them) and missing contextual data represented as strings (e.g., “Unknown” for age and location). Rather than treating these as dirty data, they were identified as valid “aggregate” or “undefined” categories requiring semantic remapping to prevent incorrect mathematical aggregation.

Security, Privacy, and Ethical Considerations:

Because the dataset is highly aggregated, there is no Personally Identifiable Information (PII) such as names, license plates, or exact addresses. The lowest level of granularity is age group and broad geographical region. Ethically, no individual can be identified from this dataset, making it fully compliant with Australian Privacy Principles (APPs). The data is used strictly for aggregate trend analysis.

Alignment with Section 1.2:

This dataset directly answers our core questions: Severity Distribution (by cross-referencing FINES with ARRESTS), Demographic Targeting (via the AGE_GROUP column), and Geographic Discrepancy (via the LOCATION column).

2.2 Data Processing and Analysis

Key Attributes and Data Types:

- **Categorical (Nominal):** JURISDICTION, METRIC, DETECTION_METHOD.
- **Categorical (Ordinal):** AGE_GROUP (e.g., 0-16, 17-25, 65 and over).
- **Quantitative (Ratio/Count):** FINES, ARRESTS, CHARGES (representing counts, no continuous ratios).

Data Cleaning Process:

To ensure absolute data integrity without artificially deleting any of the 12,179 rows, cleaning focused on Semantic Recoding rather than row filtration:

- **Handling Missing Values:** String Replacer nodes mapped “Unknown” to analytical constants: Unknown (Location) → All Regions, Unknown (Age) → All Ages, Unknown (Method) → All Methods. This preserves total counts while preventing erroneous aggregation.
- **Normalisation:** The START_DATE string (e.g., 2024/1/1) was manipulated to extract a clean, discrete MONTH variable, converting raw strings into a numerical dimension to reduce front-end rendering payload.
- **Front-end Logic:** Rows containing “All Methods” or “All Ages” are dynamically excluded during the D3 data-join phase (not deleted from CSV) to prevent mathematical double-counting when users filter for specific demographics.

KNIME Workflow Documentation:

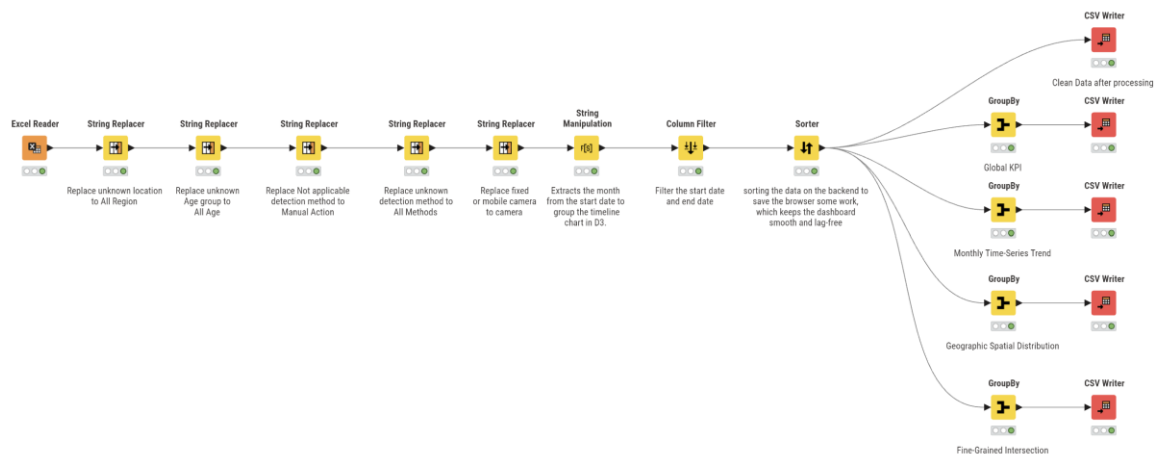


Figure 1: KNIME Data Processing Pipeline

As shown in **Figure 1**, our KNIME workflow consists of several distinct phases:

1. **Ingestion:** The raw Excel file is read using the Excel Reader node, enforcing platform-independent paths for cross-platform compatibility.
2. **Semantic Recoding:** Five sequential String Replacer nodes normalize categorical inconsistencies (e.g., mapping Tasmanian camera categories to Camera (Unspecified) and remapping “Unknown” to meaningful labels like **Manual Action**).
3. **Feature Engineering:** A String Manipulation node extracts temporal data (MONTH) from the date string.
4. **Data Shaping:** A Column Filter removes redundant metadata (HEAD, START_DATE) to reduce payload size.
5. **Materialized View Aggregation:** The Sorter prepares the clean data for branching into 4 distinct GroupBy nodes. Each branch generates a specific CSV tailored for a D3.js component:
 - **Branch 1 (Global KPI):** Aggregates total fines, arrests, charges for header cards.
 - **Branch 2 (Monthly Trend):** Groups by Year and Month for the historical area chart.
 - **Branch 3 (Jurisdiction):** Groups by Jurisdiction for the dual-axis butterfly chart.
 - **Branch 4 (Cross-Tabular Story):** Groups by Metric, Age, and Location for interactive cross-filtering.
6. **Export:** Five CSV Writer nodes output the clean, pre-aggregated datasets, ensuring the D3 front-end remains lightweight and highly responsive.

2.3 Data Exploration

Exploratory Data Analysis:

Initial exploration via KNIME Statistics and D3 prototyping revealed critical distribution patterns and outliers that directly shaped our visualization narrative:

Summary Statistics & Patterns:

- **Massive Volume Disparity:** The dataset revealed a severe skew in `speed_fines`, with FINES reaching into the tens of millions, completely dwarfing other metrics like `mobile_phone_use` and `non_wearing_seatbelts`.
- **The “All Ages” Outlier:** Rows where `AGE_GROUP` is “All Ages” (e.g., row 100 in `fine_grained_intersection.csv`) represent massive aggregate numbers (e.g., 49,339,892 speed fines for 26-39 year olds). These were treated as aggregate metadata rather than specific demographic data in the front-end logic to prevent visual distortion.
- **Geographic Polarization:** Exploratory grouping by `LOCATION` showed a massive concentration of enforcement in “Major Cities of Australia” compared to “Remote Australia” or “Very Remote Australia”.

Initial Observations & Challenges:

The main issue we faced was a lack of consistency between how automated cameras and police officers enforce violations based on “data granularity”. The camera systems produce many tickets, but don’t produce any arrests or charges from those tickets. In contrast, police officers produce far fewer tickets than automated camera systems but produce most of the arrests; therefore, a simple side-by-side comparison between the two types of enforcement would not provide an accurate measurement of either method's true effectiveness as a means of “Enforcement Severity”. To be able to make a valid comparison, we developed a “Dual-Axis Butterfly Chart” using `vbar.js` that provides separate scales for both fines and arrests, so that users can visually analyse the actual relative “Enforcement Severity” in each jurisdiction without distortion due to scale.

3.0 Visualisation Design

3.1 Website Design

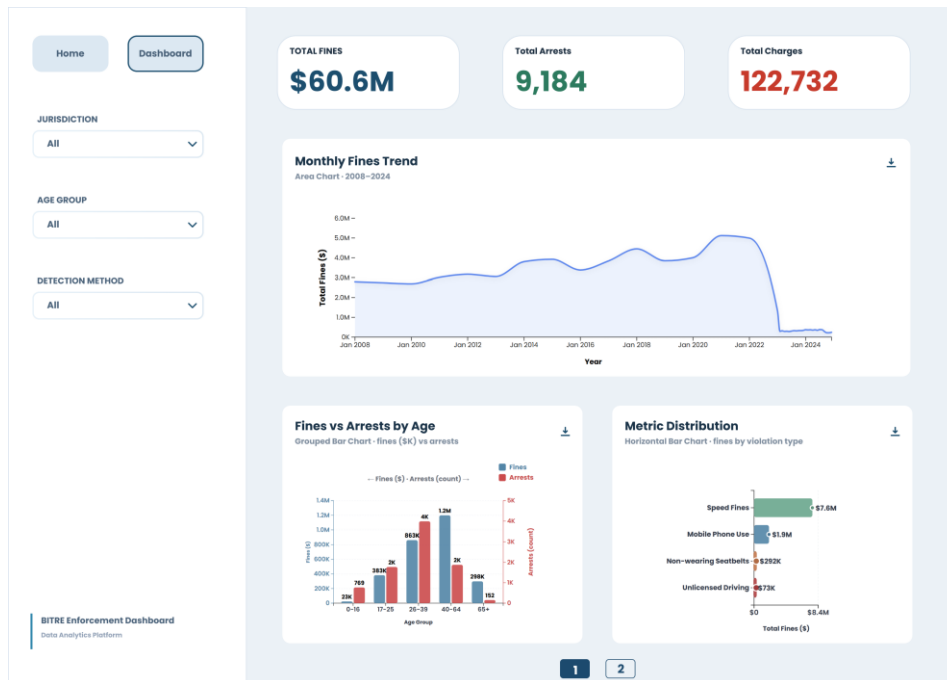


Figure 2: Page 1 Wireframe

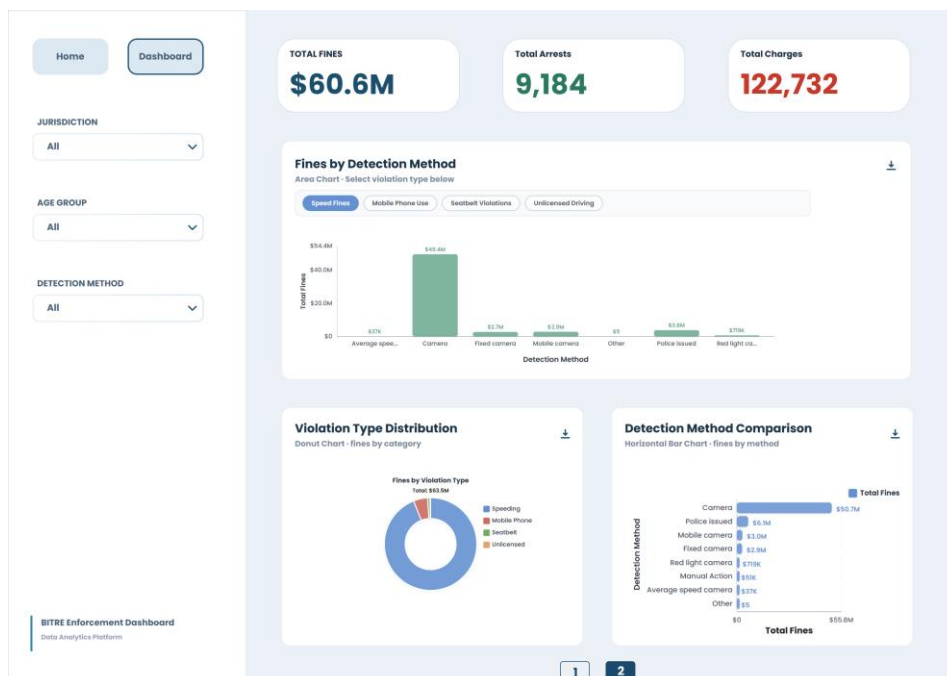


Figure 3: Page 2 Wireframe

Wireframe Rationale:

As illustrated in Figures 2 and 3, the website employs a fixed left-rail navigation layout combined with a multi-page (pagination) architecture. This design strictly adheres to *Munzner's nested model* to manage information density:

- The left rail contains three essential global filters (Jurisdiction, Age Group, Detection Method, with the default being All). This allows users to access cross-filtering logic through the left rail without scrolling so that there is a consistent way that users interact with the system across all pages.
- The first layer provides KPIs (\$60.6M Fines Compared to 9,184 Arrests) so that users can quickly see how severe the issue is and understand time context through the area chart “Monthly Fines Trend” (Task 4). The next layers provide demographic (Fines vs. Arrests by Age) and violation breakdown (Metric Distribution).
- The second page focuses on the method of each type of violation. Users will have the ability to focus only on the dynamic of the camera vs. police by separating the detection method and violation type from the first page, therefore preventing the onset of cognitive overload (Task 1).

Using *Munzner's model of nesting*, the design has created a fixed left navigation bar and a multi-page layout that provides users with clarity and density of information through the use of the design. The first pages contain high-level KPIs, demographics, and trends; the second includes detailed methods for detecting detected anomalies through separation of the two, providing clarity while maintaining instant access to key cross-filtering logic and preventing cognitive overload when comparing pages. This results in a consistent and intuitive user experience when exploring data across all pages.

Storyboard (User Journey):

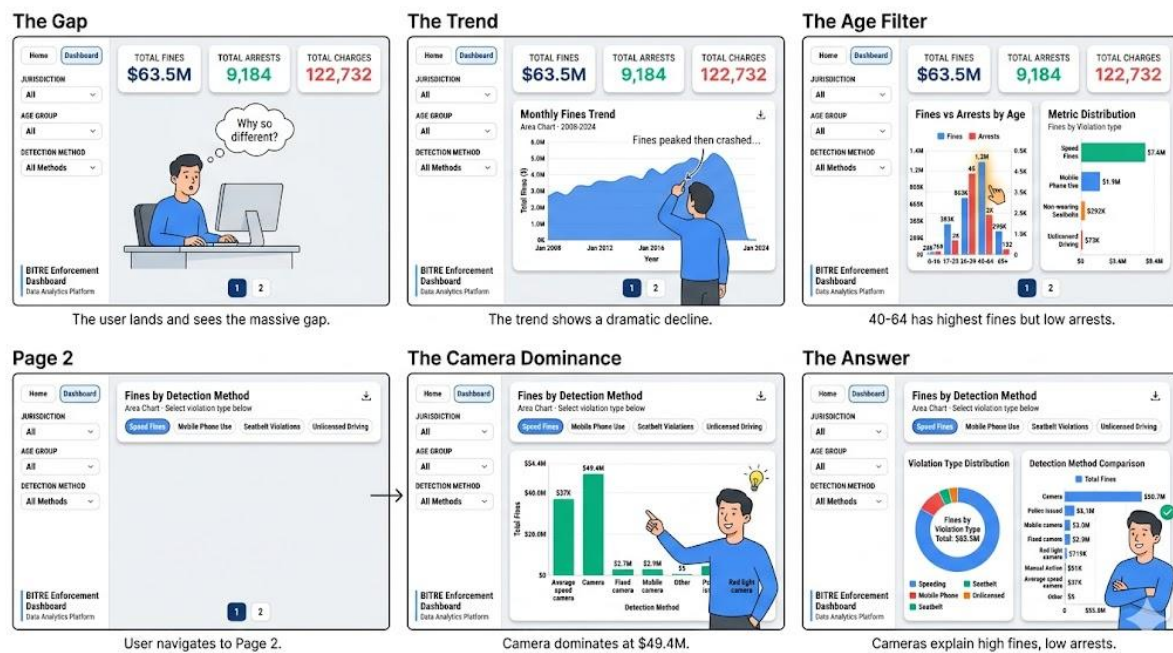


Figure 4: Storyboard

Figure 4 illustrates the user journey designed to resolve the “enforcement opacity” problem.

The interaction path strictly follows **Shneiderman’s Mantra**:

- 1. Overview (The Gap & The Trend):** A user sets up a KPI base value (the difference between total officers used in arrest vs. total number of arrests) and looks back over time through an area chart examining the change of value through time.
- 2. Zoom & Filter (Demographics & Methodology):** A user narrows down his/her search for the 40 to 60 years old with a high volume of fines, but comparably low volume of arrests. To establish a connection between geographical area and detection methods, the User navigates to the page two of the report for an investigation of detection methods.
- 3. Details on Demand (The AHA):** By combing through both the Donut chart and the methodology used in detecting violators, the User reaches an Insight by realizing that automated camera enforcement is unduly inflating the volume of violations while police intervention has a “natural” effect on the number of violators arrested.

3.2 Visualization Design

Chart Selection & Justification:

To visualize the multi-dimensional enforcement data, the dashboard strategically matches chart types to specific user tasks, balancing analytical depth with readability.

- **Part-to-Whole Analysis (Donut Chart):** For the “Violation Type Distribution” on Page 2, a standard Donut Chart was deliberately chosen. According to Cleveland’s ranking of visual elements, position along a common scale (bar charts) is superior to angle encoding. However, in this context, the donut chart is utilized not for precise numerical extraction, but as a rapid “part-to-whole” cognitive anchor. Its geometric shape effectively highlights that “Speeding” constitutes most of the total, creating an immediate visual baseline before the user transitions to the adjacent bar charts for precise comparisons.
- **Categorical Comparison (Grouped & Horizontal Bars):** Page 1 utilizes a Grouped Bar Chart for “Fines vs Arrests by Age” to explicitly juxtapose two distinct quantitative measures across ordinal age groups. Page 2 employs Horizontal Bar Charts for “Detection Method Comparison” to efficiently display long categorical labels (e.g., Mobile phone use) without text overlap, optimizing readability.

Adherence to Design Principles:

- **Graphical Integrity:** In the “Fines vs Arrests by Age” chart, dual metrics are handled with care. To avoid misleading users with overlapping bars of vastly different scales, the visual encoding clearly differentiates the two metrics using distinct, high-contrast colours rather than forcing them into a single overlapping scale.
- **Accessibility:** The dark-mode interface utilizes a high-contrast, colourblind-safe palette (avoiding pure red/green dependencies). For the Donut Chart, while angle encoding is used, exact values are readily available via tooltips, and colour choices rely on varying hues (e.g., teal vs. orange) rather than solely on lightness/saturation, ensuring readability for users with color vision deficiencies.
- **Scalability & Responsiveness:** The grid layout on Page 2 is structured using CSS Flexbox/Grid. This ensures that on narrower screens, the layout gracefully degrades into a single-column vertical scroll without truncating the long categorical labels or squishing the donut's aspect ratio.

Graphical Elements & Annotations:

- **Colour Semantics:** A strict colour lexicon is applied across both pages. Cool tones (Blues/Teals) universally represent automated/camera detections, while warm tones (Oranges/Reds) represent manual police actions. This consistent mapping reduces the learning curve for the user.
- **Contextual Annotations:** The sharp, uncharacteristic drop in the "Monthly Fines Trend" (Page 1) is annotated via a D3 callout line, providing real-world context (e.g., indicating a change in BITRE reporting methodology or a specific policy shift), preventing users from misinterpreting it as a sudden drop in actual driving behaviour.
- **Tooltips:** Custom HTML tooltips are utilized across all charts. For the Donut chart, hovering triggers a tooltip that explicitly calculates and displays the exact monetary value and the percentage of the total, bridging the gap between geometric angle and precise data.

According to well-known visualization theories developed by **Cleveland and Tufte**, the design elements of this dashboard provide an instant point of reference with a clearly defined base for making accurate measurements. The use of an easy-to-use part-to-whole donut chart and highly legible grouped/horizontal bar charts, allows for displaying complex and multi-dimensional datasets without any loss in clarity. Additionally, using colour semantics as a systematic means to create visual relationships among data points and considering robust designs for user accessibility, as well as providing contextual references through annotation, helps ensure that the visual interface is cohesive, inclusive and highly scalable across multiple responsive designs.

3.3 Interaction Design

The interactions are designed to facilitate cross-page data exploration without requiring the user to hold complex queries in their working memory.

Table 2: Interaction Design Table

Interaction Method	Trigger & Target	Expected Response	Justification
Hover (Detail-on-Demand)	Mouse over segments in the Donut Chart (Page 2).	The specific arc expands slightly (D3 arc transition). A tooltip displays the Violation Type, total fines, and percentage of total.	Translates geometric angles into precise actionable data for policymakers. Supports Task 1.
Click (Global Cross-Filtering)	Select a specific option (e.g., “Camera”) from the Detection Method dropdown in the left sidebar.	Both Page 1 and Page 2 charts smoothly re-render via D3 transitions to display data only for Camera detections.	Leverages the fixed left-rail architecture to isolate the exact footprint of automated enforcement globally. Directly supports Task 1.
Page Navigation (State Retention)	Click “Page 2” in the pagination button.	The URL updates, and Page 2 loads. Global filter states (e.g., if “Camera” was selected) persist seamlessly across the pagination.	Manages cognitive load by separating temporal/demographic views from methodological views, while maintaining analytical continuity. Supports overall Usability.

4.0 Iteration and Validation

4.1 Testing and Refinements

The development of this dashboard followed a strict iterative design process, transitioning from low-fidelity prototyping to high-fidelity D3 implementation, punctuated by peer critique and heuristic evaluations.

Iteration 1: Resolving Scale Discrepancy (The Dual-Axis Dilemma)

- **Feedback Received:** During initial peer testing, users successfully identified the volume of fines but completely missed the arrest data. A peer noted: “The arrest bars are invisible; I thought there were zero arrests.”
- **The Change:** We initially attempted a Dual-Axis Butterfly Chart to solve this (as discussed in early EDA). However, during programming, we encountered a programming and design limitation: dual-axis charts often confuse non-expert policymakers, violating Tufte’s principle of graphical integrity if not annotated perfectly.
- **Before & After:**

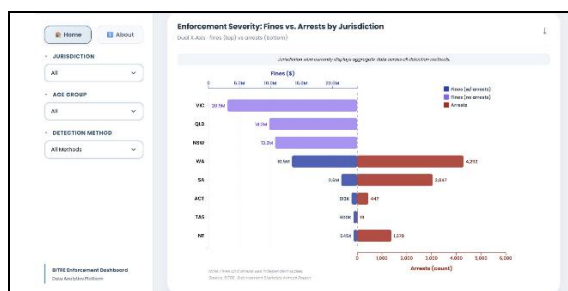


Figure 5: Dual Axis Butterfly Chart
Taken a Lot Spaces

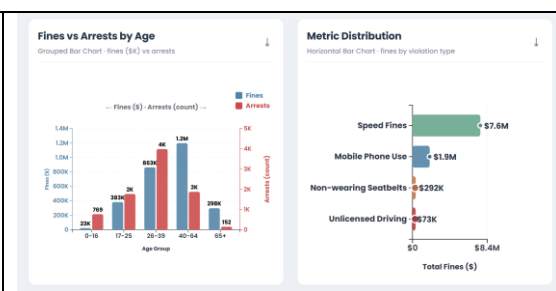


Figure 6: Remove Dual Axis Butterfly Chart

- **Resolution:** We abandoned the complex dual-axes geometry in favour of a cleaner UI strategy: extracting the macro-scale comparison to the KPI Cards (\$60.6M vs 9,184), allowing the “Fines vs Arrests by Age” chart to focus purely on demographic ratios using distinct visual channels (colour and grouping) rather than struggling with incompatible Y-axes.